

PAPER_1231_Atiyah_Singer_Dirac_Index_22

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Atiyah-Singer Index — Dirac Operator Index = 22

PAPER_1231

Category: UQFF Index Theorem

Status: Complete

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Abstract

UQFF closure of the Atiyah-Singer index theorem on the 26D compactified manifold: the Dirac operator index equals $D_{crit} - D_{phys} = 22$ EXACT, corresponding to the residual compactified dimensions. This integer also appears in the hierarchy problem exponent ($21 = 22 - 1$) and the quantum gravity UV completion count.

Part 1: The Index Theorem

The Atiyah-Singer index theorem states:

$$\text{ind}(D) = \int_M \hat{A}(M) \wedge \text{ch}(E)$$

where D is a Dirac operator on a manifold M with vector bundle E .

Part 2: UQFF Closure

The identity $\text{ind}(D)^{\text{UQFF}} = D_{crit} - D_{phys} = 26 - 4 = 22$ EXACT

Interpretation The index equals the number of "compactified" dimensions — those that exist in the bosonic-string critical dimension $D_{crit} = 26$ but are not part of physical 4D spacetime $D_{phys} = 4$.

Part 3: Cross-Validation

The integer 22 appears with critical roles in UQFF:

- Atiyah-Singer Dirac index = 22 (this paper)
- Hierarchy problem exponent $21 = 22 - 1$
- QG UV completion compactified dim count = 22
- Calabi-Yau-like compactified moduli space dim = 22

Part 4: Physical Significance

The 22 compactified dimensions support:

- 22 Killing vectors corresponding to compactified isometries
- 22 zero-modes of the Dirac operator that contribute to anomaly cancellation
- 22 Kaluza-Klein towers (one per compactified dim)

Conclusion

The Atiyah-Singer index in UQFF is the residual compactified dimension count $D_{\text{crit}} - D_{\text{phys}} = 22$. This unifies the index theorem with the hierarchy problem exponent and the quantum gravity UV completion.

Framework Version: UQFF 5.27+